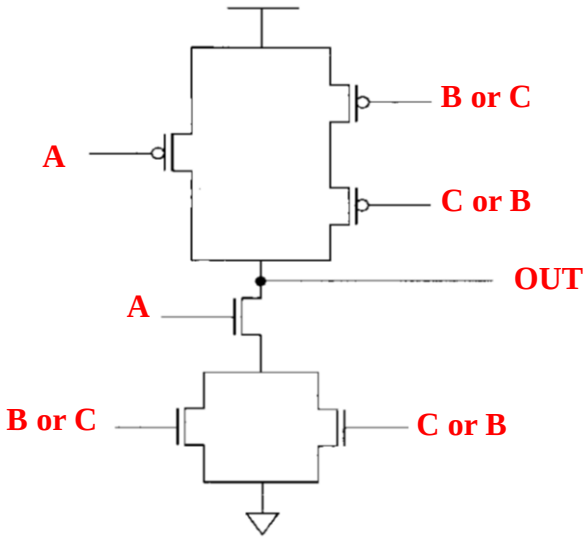
# CSE1100 – Introduction to Computer Engineering
# Homework 2 (Due 31/10/2017 17:00)

1) Read Chapters 3.1, 3.2, and 3.3 from the book.

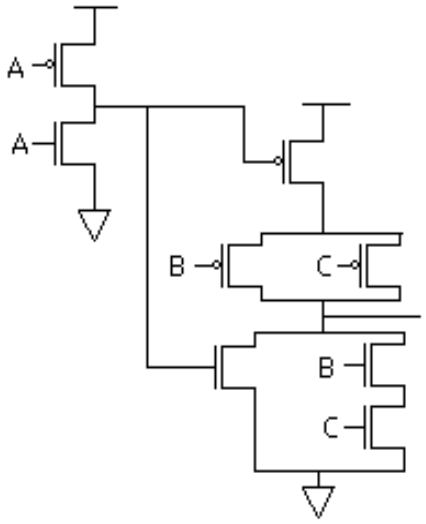
2) The transistor-level circuit below implements the logic equation given below. Give the complete truth table for the given logic equation, and label the inputs to all the transistors.

Y = NOT (A AND (B OR C))




| A | B | C | OUT |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 0 |

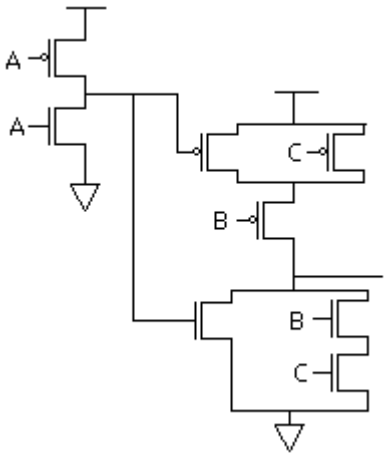
3) Give the complete truth table for the transistor-level circuit below, and give the logic equation that implements this circuit.




| A | B | C | OUT |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

(AB'C') OR (AB'C) OR (ABC')

4) What is wrong with the transistor-level circuit below? Explain.




If you input A=0, B=0, and C=0 or A=1, B=1, C=1 there will be a path from power to ground.

5) Given the truth table below,

| A | B | C | X |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

a. Draw the corresponding transistor-level circuit.

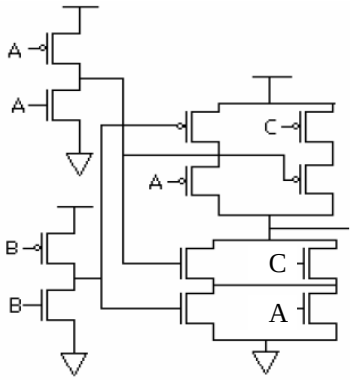
b. Draw the corresponding gate-level logic circuit, using only NOT, AND, and OR gates. Also, give the logic equation.

c. How can you implement the same truth table using only NAND gates? Draw the gate-level logic circuit with NAND gates.

d. How many transistors are needed for circuits in (a) and (b)?

a) If we write the logic equation for the truth table as:

$F = (A' \cdot B \cdot C') + (A' \cdot B \cdot C) + (A \cdot B' \cdot C') + (A \cdot B \cdot C') == A' \cdot B + A \cdot C'$



Explanation:

You use the upper part of a transistor level circuit to set the 1's and the lower part to set the 0's.

In the upper part we have:

- A' and B in series for AND,
- A and C' in series for AND, and
- They are in parallel (for the OR part)

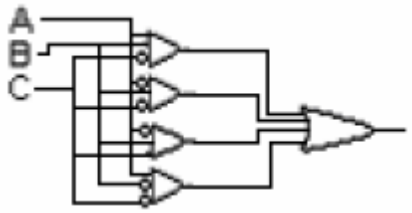
The variables enter negated into the p-type MOS transistor, because the p-type MOS transistor will turn ON for input value 0.

Then, the lower part must implement the 0's. Recalling the De Morgan Theorem, by negating F, AND becomes OR and vice versa. Applying to our function, we'll get:

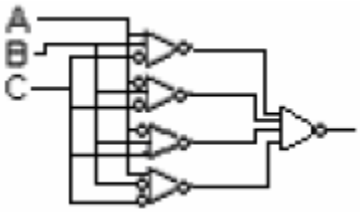
$$F' = (A+B') \cdot (A'+C)$$

Therefore, in the lower part we have A and B' in parallel and again A' and C in parallel, while these two are in series. The variables does not enter negated into the n-type MOS transistor because n-type MOS transistor turn ON for input value 1.

b) $(A' \cdot B \cdot C') + (A' \cdot B \cdot C) + (A \cdot B' \cdot C') + (A \cdot B \cdot C')$



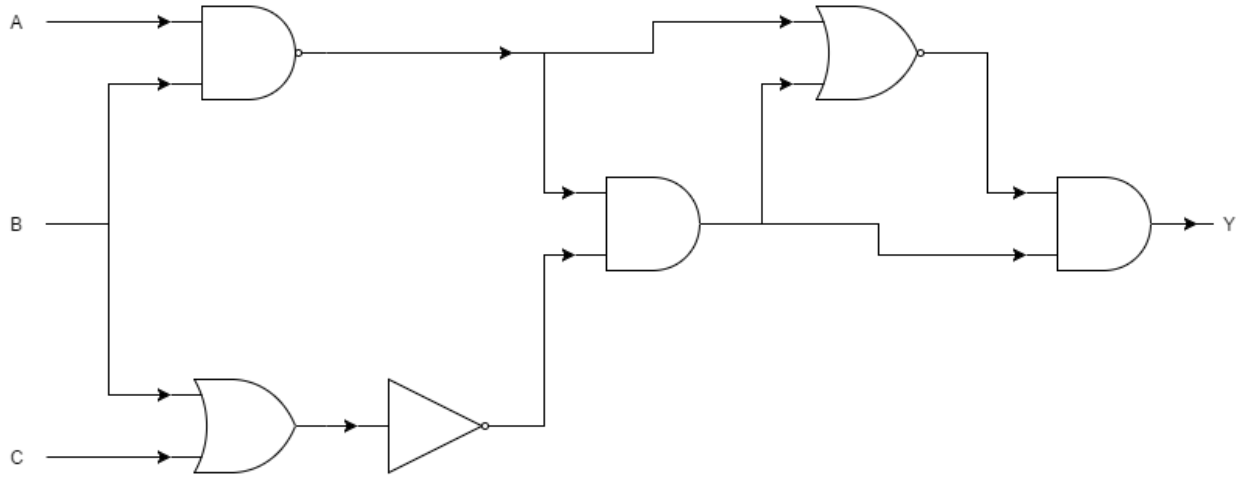

c)




d) (a) 12 transistors
(b) 54 transistors (12 transistors for 6 NOT gates (6*2), 32 transistors for 4 3-input AND gates, and 10 for 4-input OR gate → 12+32+10=54)

NOTE: The circuits may be drawn in a different way. This is one of possible solutions.

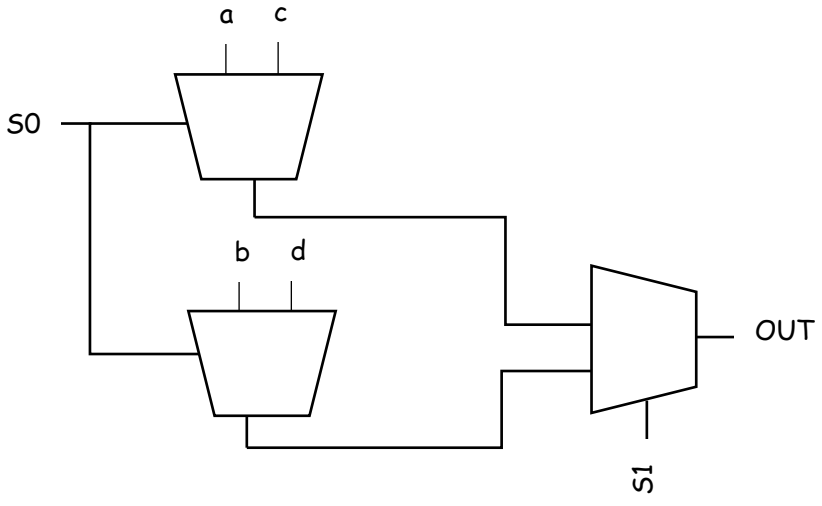
6) Given the logic circuit below, write the corresponding logic equation, and fill in the truth table for the output value Y.




7) Implement a 4-to-1 multiplexer using only 2-to-1 multiplexers. Assume that the output (OUT) will be as in the following table according to the control signals (S1 and S0). (a,b,c, and d are the inputs to the 4-to-1 multiplexer). Make sure that all of the terminals are properly connected.

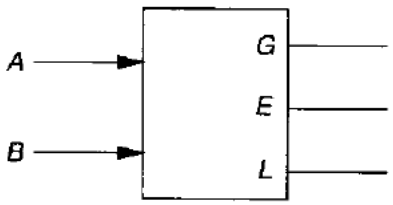
**Hint:** Three 2x1 multiplexers are enough

| **S0** | **S1** | **OUT** |
|---|---|---|
| 0 | 0 | a |
| 0 | 1 | b |
| 1 | 0 | c |
| 1 | 1 | d |



8) A comparator circuit has two 1-bit inputs, A and B, and three 1-bit outputs, G (greater), E (equal), and L (less than) as given in the following figure.

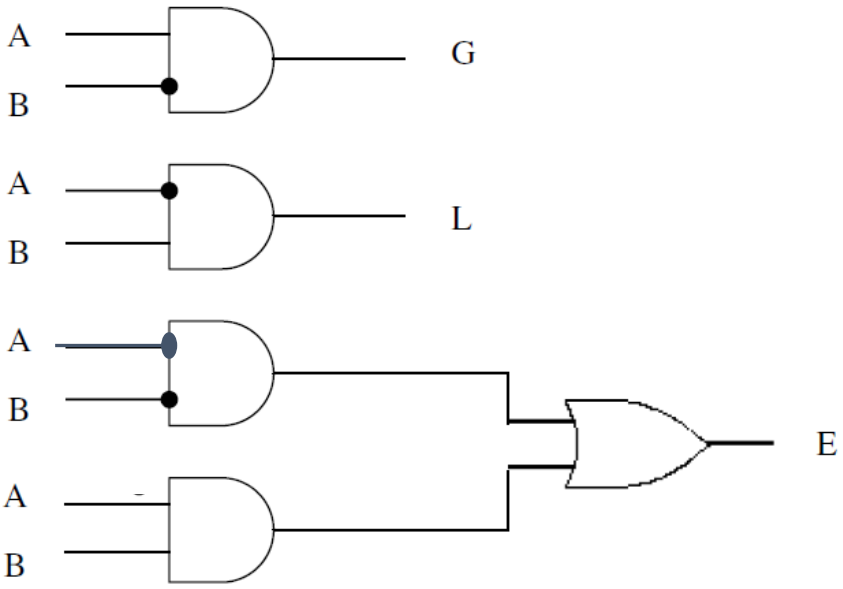
| G is 1 if $A > B$ | E is 1 if $A = B$ | L is 1 if $A < B$ |
|---|---|---|
| 0 otherwise | 0 otherwise | 0 otherwise |




a) Draw the truth table for this 1-bit comparator.

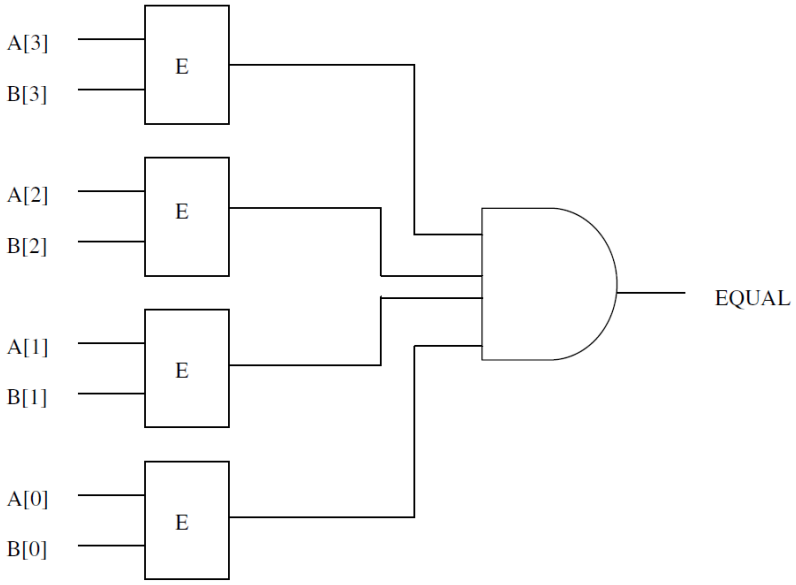
| A | B | G | E | L |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 1 | 0 |

b) Implement G, E and L for a 1-bit comparator using AND, OR, and NOT gates.

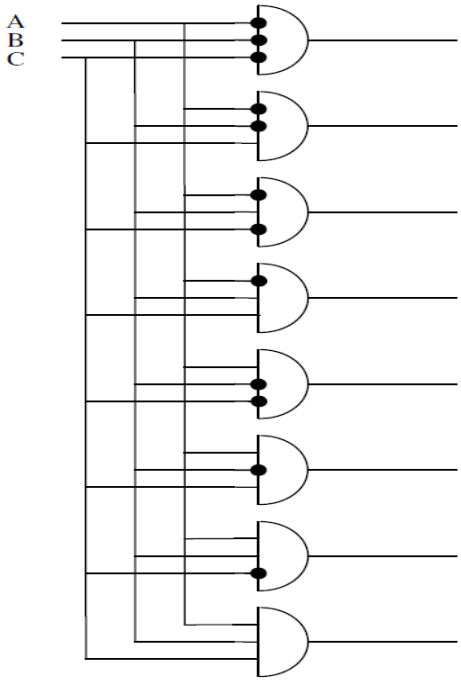



c) Using the one-bit comparator as a basic building block, construct a four-bit equality checker, such that output EQUAL is 1 if all bits of A are equal to all bits of B, 0 otherwise.

**Hint:** You may not need to use all 12 inputs.




9) Design a 3-to-8 decoder with inputs (A, B, C) using only AND gates.




You should submit your hand-written homework by the due date to your assistant Ozan Neli (Room MB650).